Computer Science 101: Introduction to Artificial Intelligence

Lecture 2: Artificial Neural Networks

1. Biological inspiration: Neurons, synapses, and action potentials.
2. The Perceptron: Frank Rosenblatt (1958).
3. Multi-Layer Perceptrons and Backpropagation.

Activation Functions:
- Sigmoid: f(x) = 1 / (1 + exp(-x))
- ReLU: f(x) = max(0, x)
- Softmax: Used for categorical probability distributions.

Optimization:
Gradient descent updates weights iteratively in the direction of steepest loss decrease.